Kummer lines
Doubling
Diff. add.
Conclusion

Arithmetic of Kummer lines

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Motivation

Figure: An RFID tag

- ECDSA and ECDH rely on the scalar product of an elliptic curve, we'd like to improve that.
- Having efficient isogeny formulas is useful for post-quantum cryptography.

Outline

1. Kummer lines
   - Montgomery ladder
   - Definition
2. Doubling
   - Isogenies
   - Doubling formulas
   - Hybrid ladder
3. Differential addition
   - Half differential addition formulas
   - Half ladder

Kummer lines

Elliptic curves (char $k \neq 2, 3$)

- Short Weierstrass (general case):

$$E : y^2 = x^3 + ax + b$$

- Montgomery curves:

$$E : y^2 = x(x^2 + \mathcal{A}x + 1)$$

- How to compute efficiently $n \cdot P = P + \cdots + P$?

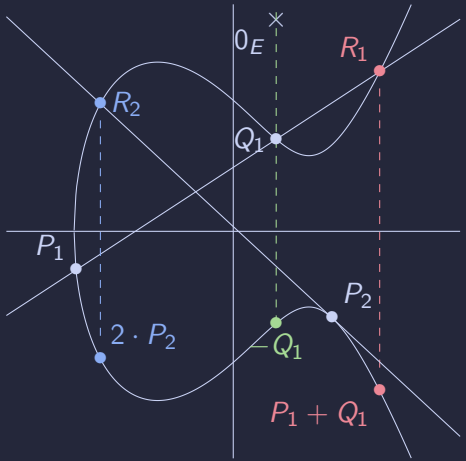

Figure: An elliptic curve

x-only arithmetic

From $(x(P), y(P))$ and $(x(Q), y(Q))$, we can compute $(x(P+Q), y(P+Q))$.
What if we only keep $x(P)$ and $x(Q)$?

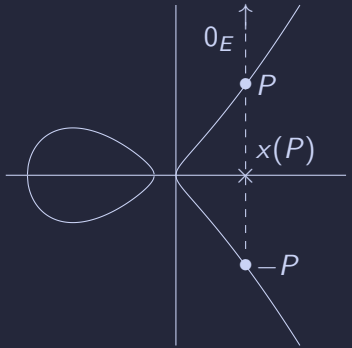

Figure: A Montgomery curve

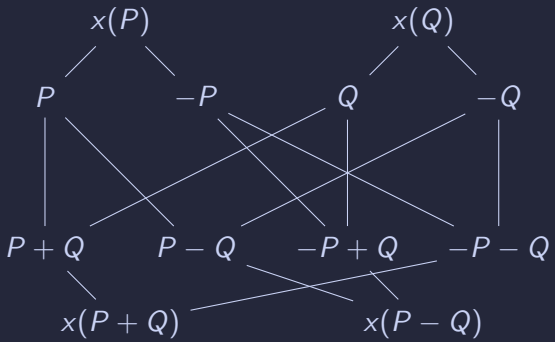

Figure: Two possible choices

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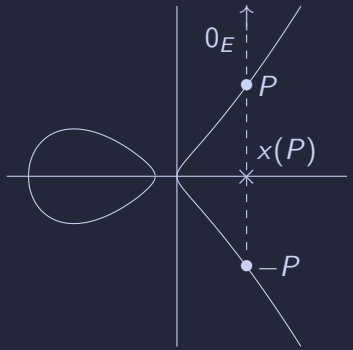

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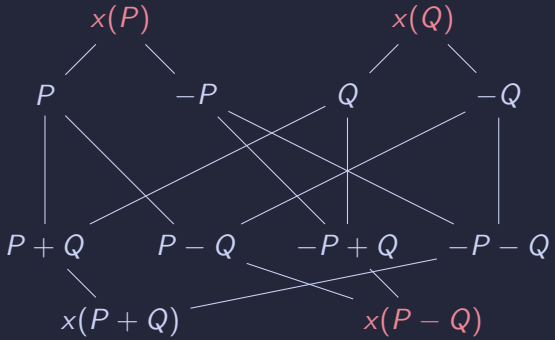

Figure: Two possible choices

Differential addition: knowing $x(P)$, $x(Q)$, $x(P-Q)$, we can compute $x(P+Q)$.

Arithmetic on $y^2 = x(x^2 + \mathcal{A}x + 1)$[1]

Differential addition ($3M + 2S$)

$u := (X_P + Z_P)(X_Q - Z_Q)$, $v := (X_P - Z_P)(X_Q + Z_Q)$.

$$X_{P+Q} = (u+v)^2, \quad Z_{P+Q} = \frac{X_{P-Q}}{Z_{P-Q}}(u-v)^2.$$

Doubling ($2M + 2S + 1m_0$, $d = \frac{\mathcal{A}+2}{4}$)

$u := (X_P + Z_P)^2$, $v := (X_P - Z_P)^2$, $t := u - v$.

$$X_{2.P} = uv, \quad Z_{2.P} = t(v + dt).$$

[1] P. L. Montgomery. "Speeding the Pollard and elliptic curve methods of factorization". In: *Mathematics of Computation* 48 (1987), pp. 243–264.

Algorithm 1: Montgomery ladder step

Input: $R = m \cdot P$, $S = (m+1) \cdot P$, b a bit
Output: $(2 \cdot R, R+S)$ if $b = 0$; $(R+S, 2 \cdot S)$ if $b = 1$.
Data: The point P

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def xDBLADD(R, S, b):
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$n = 11 = \overline{1011}^2$

0_E ——— P

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$$\begin{tikzcd} 0_E \arrow[r, dash] \arrow[d] & P \arrow[d] \arrow[dl] \\ P \arrow[r, dash] & 2P \end{tikzcd}$$

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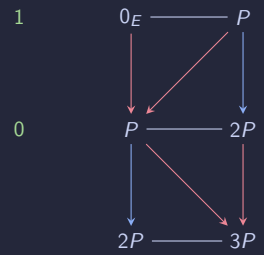

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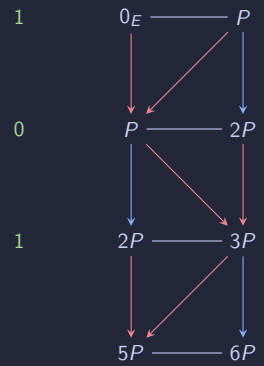

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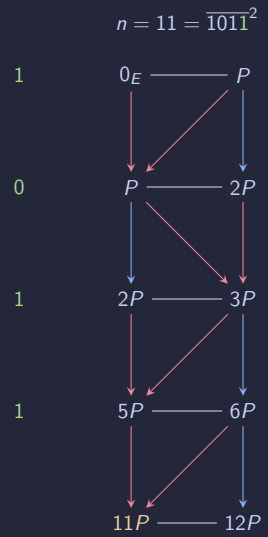


Figure: Montgomery ladder

Kummer line model of a Montgomery curve

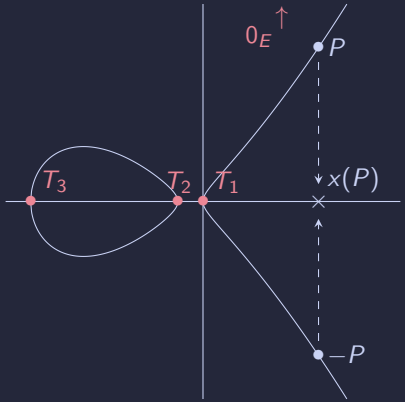
$$E : BY^2Z = X(X^2 + \mathcal{A}XZ + Z^2)$$

$$x : E \to \mathbb{P}^1, \begin{cases} 0_E & \mapsto \infty := (1:0), \\ (X:Y:Z) & \mapsto \frac{X}{Z} := (X:Z). \end{cases}$$

Ramification

Since $x(P) = x(-P)$:

- $\#x^{-1}(x(P)) = 1$ when $P = -P$
  $\rightsquigarrow$ this is 2-torsion $E[2]$: 4 elements;
- $\#x^{-1}(x(P)) = 2$ otherwise.



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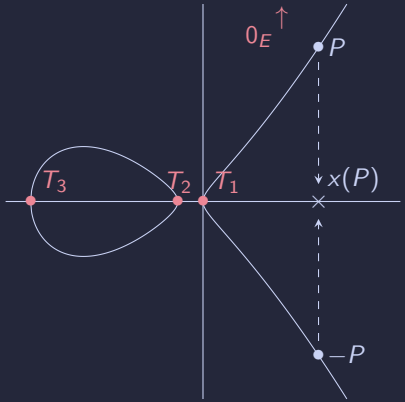

Figure: Ramification on the curve

$$0_E = (1:0)^*, \quad T_1 = (0:1), \quad T_2 = (\alpha:1), \quad T_3 = (1:\alpha).$$

Kummer line: general definition

Kummer line

A Kummer line of an elliptic curve E is:

1. A degree 2 covering $\pi : E \to \mathbb{P}^1$:

$$\pi^{-1}(\pi(P)) = \{-P, P\}.$$

2. 4 ramification points, which correspond to the 2-torsion:

$$\pi^{-1}(\pi(T)) = \{T\} \text{ for } T \in E[2].$$

A map between Kummer lines $\varphi : \mathbb{P}^1 \to \mathbb{P}^1$ maps the ramification points to ramification points.

Examples

Montgomery model $M(a:b)$: $y^2 = x(x - a/b)(x - b/a)$, $a/b \in \overline{k}$

$$\pi : (x, y) \mapsto (x : 1),\ 0_E \mapsto (1 : 0).$$

$$0_E = (1 : 0)^*, \quad T_1 = (0 : 1), \quad T_2 = (a : b), \quad T_3 = (b : a).$$

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Theta squared model $\theta_s(a:b)$: $y^2 = x(x - a/b)(x - b/a)$, $a/b \in k$

$$\pi : (X:Y:Z) \mapsto (aX - bZ : bX - aZ),\ 0_E \mapsto (a:b).$$

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Theta model $\theta(a:b)$

$$0_E = (a:b)^*, \quad T_1 = (-a:b), \quad T_2 = (b:a), \quad T_3 = (-b:a).$$

Arithmetic on $\theta_s(a:b)$[2]

Differential addition ($3M + 2S + 1m_0$ | Montgomery: $3M + 2S$)

$u := (X_P + Z_P)(X_Q + Z_Q)$, $v := \frac{a+b}{a-b}(X_P - Z_P)(X_Q - Z_Q)$.

$$X_{P+Q} = (u+v)^2, \quad Z_{P+Q} = \frac{X_{P-Q}}{Z_{P-Q}}(u-v)^2.$$

Doubling ($4S + 2m_0$ | Montgomery: $2M + 2S + 1m_0$)

$u := (X_P + Z_P)$, $v := \frac{a+b}{a-b}(X_P - Z_P)$.

$$X_{2.P} = (u+v)^2, \quad Z_{2.P} = \frac{a}{b}(u-v)^2.$$

[2] P. Gaudry and D. Lubicz. "The arithmetic of characteristic 2 Kummer surfaces and of elliptic Kummer lines". In: *Finite Fields & Their Appl.* 15.2 (2009), pp. 246–260.

Kummer lines
Doubling
Isogenies
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Conclusion

Doubling

Isogenies (between elliptic curves): definition

Definition

Surjective homomorphism $f : E \to E'$ with finite kernel.
$\rightsquigarrow$ Induces a field extension $f^* : k(E') \to k(E)$: separable, $\deg f$, $\deg_s f$, ...

Some examples:

- the multiplication by $m \in \mathbb{Z}$: $[m]_E : P \in E \mapsto m \cdot P$;
- $f : (E : y^2 = x(x^2 + 3x + 1)) \to (E' : y^2 = (x-2)(x+2)(x+3))$ where

$$f : (x, y) \mapsto \left(\frac{x^2+1}{x}, y\frac{x^2-1}{x^2} \right).$$

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At the Kummer line level, with the x-coordinate model, the induced map is

$$\varphi : (X : Z) \mapsto (X^2 + Z^2 : XZ).$$

Isogenies: cryptography

Why are we interested in these objects?

- Natural to study the morphisms of elliptic curves;
- More importantly, post-quantum cryptography candidates.

$$\begin{tikzcd} E \arrow[r, "f_A"] \arrow[d, "f_B"'] & E_A \arrow[d, "f_{AB}"] \\ E_B \arrow[r, "f_{BA}"'] & E_{AB} \end{tikzcd}$$

Figure: SIDH protocol (attacked in 2022)

$$\begin{tikzcd} E_2^2 \arrow[r, "\alpha_2"] & E_2^2 \\ E_1^2 \arrow[u, "\Sigma"] \arrow[r, "\alpha_1"'] & E_1^2 \arrow[u, "\Sigma"'] \end{tikzcd}$$

$$\rightsquigarrow F = \begin{pmatrix} \alpha_1 & \widetilde{\Sigma} \\ -\Sigma & \alpha_2 \end{pmatrix} \in \mathrm{End}(E_1^2 \times E_2^2)$$

Figure: SQISignHD protocol

Isogenies: dual

Consider d such that $d \nmid \operatorname{char} k$, and f an isogeny with kernel $\ker f = \langle T \rangle \simeq \mathbb{Z}/d\mathbb{Z}$. There is a dual isogeny $\widetilde{f} : E' \to E$ such that

$$\widetilde{f} \circ f = [d]_E \quad \text{and} \quad f \circ \widetilde{f} = [d]_{E'}.$$

In particular, when $d = 2$, we can factor the multiplication by 2, giving fast doubling formulas.

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Goal

Are there fast 2-isogeny formulas between models of Kummer lines?

A methodology to find 2-isogenies formulas[3]

If $f : E \to E'$ is a 2-isogeny and $\ker f = \{0_E, T\}$, then $f(\cdot + T) = f$.
At the Kummer line level $\pi : E \to \mathbb{P}^1$, translation by T descends to $\mathbb{P}^1$:
$t_T : \mathbb{P}^1 \to \mathbb{P}^1$.

1. t_T is a homography, and can be computed from the ramification points of π.

[3] D. Robert and N. S. "Computing 2-isogenies between Kummer lines". In: *Communications in Cryptology* 1.1 (2024), p. 26

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2. t_T acts on X^2, Z^2, XZ: helps to build invariants u, v of degree 2 by t_T:

$$\varphi : (X : Z) \mapsto (u(X : Z) : v(X : Z)), \text{ such that } \varphi \circ t_T = \varphi.$$

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3. We can derive the ramification on the right model from the ramification of π, and shape it with a homography.

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Some old formulas, in a unified manner (1/2)

Starting from $M(a:b)$ associated to $y^2 = x(x-a/b)(x-b/a)$ with ramification

$$0_E = (1:0)^*, \quad T_1 = (0:1), \quad T_2 = (a:b), \quad T_3 = (b:a).$$

Kernel T_1 (Hisil, Renes, 2019)

Assume $a/b \in k$, set $(a':b') = (a+b : a-b)$. Then $\varphi : M(a:b) \to \theta_s(b':a')$:

$$\varphi : (X:Z) \mapsto \left(b'(X+Z)^2 : a'(X-Z)^2\right).$$

The dual is $\widetilde{\varphi} : \theta_s(b':a') \to \theta_s(a:b)$:

$$\widetilde{\varphi} : (X:Z) \mapsto \left(b(X+Z)^2 : a(X-Z)^2\right).$$

Some old formulas, in a unified manner (2/2)

$$0_E = (1:0)^*, \quad T_1 = (0:1), \quad T_2 = (a:b), \quad T_3 = (b:a).$$

Kernel T_2 (Renes, 2018, $T_2' = (A:B)$ rational above T_2)

Set $(A':B') = (A+B : A-B)$, then $\varphi : M(a:b) \to \theta_s(A^2 : B^2)$:

$$\varphi : (X:Z) \mapsto \left((B'(X+Z) + A'(X-Z))^2 : (B'(X+Z) - A'(X-Z))^2\right).$$

Kernel T_1 (de Feo, Jao, Plut, 2014, $\widetilde{T_1} = (r:s)$ rational above $T_1' = (1:1)$)

Set $(\gamma:\delta) = (4rs : (r-s)^2)$, then $\varphi : M(a:b) \to M(-\gamma : 4\delta)$:

$$\varphi : (X:Z) \mapsto \left(\gamma(X-Z)^2 : 4\delta XZ\right).$$

Even more formulas, some of which are new: Legendre to Montgomery, theta to Montgomery, theta to theta, ...

Doubling up to a 2-torsion point

Factoring doubling: $\widetilde{\varphi} \circ \varphi : M(a : b) \to \theta_s(a : b)$.

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2. the map $\overline{\psi} : \theta_s(a:b) \to M(a:b)$, $(X:Z) \mapsto (X:Z)$ is a bijection, sending P_θ to $P_M + T$ where T is a 2-torsion point.

$$\begin{array}{ccccc} M(a:b) & \xrightarrow{\widetilde{\varphi}\circ\varphi} & \theta_s(a:b) & \xrightarrow{\overline{\psi}} & M(a:b) \\ P_M & & \longmapsto & & 2 \cdot P_M + T \end{array}$$

Figure: Doubling up to a 2-torsion point

$\texttt{DoublingTranslation}(P) := 2 \cdot P + T$ ($4S + 2m_0$ compared to $2M + 2S + 1m_0$)

Algorithm 2: Hybrid ladder step

Input: $R, S \in \{P, P+T\}$, b a bit

Output: $(2 \cdot R + T, R + S)$ if $b = 0$
$(R + S, 2 \cdot S + T)$ if $b = 1$

Data: The points P and $P + T$

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def xDBLADD(R, S, b):
    Q ← R − S;                          // pre-computed
    if b = 0 then
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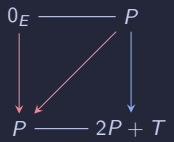

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def xDBLADD(R, S, b):
    Q ← R − S;                          // pre-computed
    if b = 0 then
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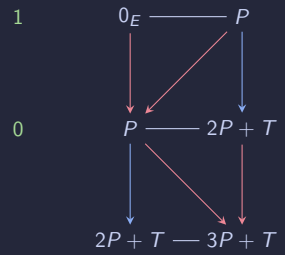

Figure: Hybrid ladder

Algorithm 2: Hybrid ladder step

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$(R + S, 2 \cdot S + T)$ if $b = 1$

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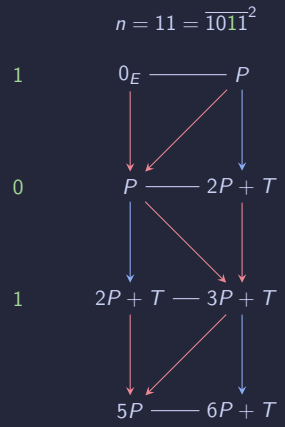

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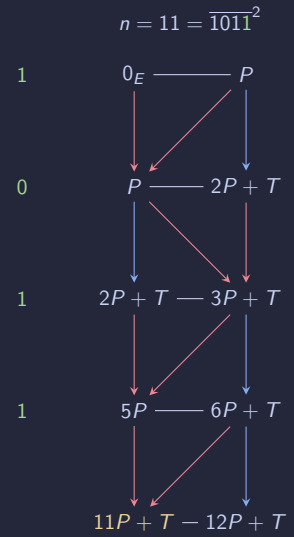

Figure: Hybrid ladder

Hybrid ladder: computational cost

Operation	Cost on $M(a:b)$	Cost on $\theta_s(a:b)$
Diff. Add.	$2M + 2S + 1m$	$2M + 2S + 1m + 1m_0$
Doubling	$2M + 2S + 1m_0$	$4S + 2m_0$

Table: Cost of operations on two models

Variant	Cost per bit
Montgomery ladder	$4M + 4S + 1m + 1m_0$
Theta ladder	$2M + 6S + 1m + 3m_0$
Hybrid ladder, our contribution	$2M + 6S + 1m + 2m_0$

Table: Comparison of different variants

When $S = 0.66M$, we need $1m_0 \leq 0.66M$ for the hybrid ladder to be faster than the Montgomery ladder.

Differential addition

The differential addition isogeny

Starting point

Differential addition formulas come from an isogeny on $E \times E$!

$$\begin{aligned} F : E \times E &\to E \times E \\ (P, Q) &\mapsto (P + Q, P - Q) \end{aligned}$$

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$$(P, Q) \mapsto (P + Q, P - Q)$$

$$\begin{tikzcd}
E \arrow[dr, "\varphi"] \arrow[dd, "{[2]_E}"'] & \\
& E' \arrow[dl, "\widetilde{\varphi}"] \\
E &
\end{tikzcd}$$

Figure: Factoring doubling

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$$\Phi : E \times E \to E' \times E'$$
$$(P, Q) \mapsto (\varphi(P), \varphi(Q))$$

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$$\begin{tikzcd} E \times E \arrow[r, "\Phi"] \arrow[d, "F"'] & E' \times E' \arrow[dl, dashed, "?"] \\ E \times E & \end{tikzcd}$$

Figure: Factoring diff. add. ?

$$\begin{tikzcd} E \arrow[dr, "\varphi"] \arrow[dd, "{[2]_E}"'] & \\ & E' \arrow[dl, "\widetilde{\varphi}"] \\ E & \end{tikzcd}$$

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$$\begin{tikzcd} E \times E \arrow[r, "\Phi"] \arrow[d, "F"'] & E' \times E' \arrow[d, dashed] \\ E \times E \arrow[r, dashed] & A \end{tikzcd}$$

Figure: Factoring diff. add. ? Not as easy

$$\begin{tikzcd} E \arrow[dr, "\varphi"] \arrow[dd, "{[2]_E}"'] & \\ & E' \arrow[dl, "\widetilde{\varphi}"] \\ E & \end{tikzcd}$$

Figure: Factoring doubling

Half differential addition

$$F : (P, Q) \mapsto (P + Q, P - Q), \quad \Phi : (P, Q) \mapsto (\varphi(P), \varphi(Q)).$$

Definition

Half differential addition formulas relative to φ are formulas such that given $\varphi(P)$, $\varphi(Q)$ and $P - Q$, can compute $P + Q$ on the Kummer line.

Notation

- $P + Q = \texttt{HalfDiffAdd}_\varphi(\varphi(P), \varphi(Q), P - Q)$;
- For consistency, $2 \cdot P = \texttt{HalfDouble}_\varphi(\varphi(P))$ $(= \widetilde{\varphi}(\varphi(P)))$.

Example on $\theta(a:b)$[4]

$$0_E = (a:b)^*, \quad T_1 = (-a:b), \quad T_2 = (b:a), \quad T_3 = (-b:a).$$

Set $(A^2 : B^2) := (a^2 + b^2 : a^2 - b^2)$, assume $A/B \in k$. The 2-isogeny considered is:

$$\varphi : (X:Z) \in \theta(a:b) \mapsto \big(B(X^2+Z^2) : A(X^2-Z^2)\big) \in \theta(A:B)$$

[4] D. Robert and N. S. "Halving Differential Additions on Kummer Lines". In: *EUROCRYPT 2025, Part VI*. vol. 15606, pp. 416–445

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$\texttt{HalfDiffAdd}_\varphi$ $(4M \mid$ Full diff. add.: $3M + 4S + 1m_0)$

$$(X_{P+Q}X_{P-Q} : Z_{P+Q}Z_{P-Q}) = \begin{pmatrix} X_{\varphi(P)}X_{\varphi(Q)} + Z_{\varphi(P)}Z_{\varphi(Q)} \\ X_{\varphi(P)}X_{\varphi(Q)} - Z_{\varphi(P)}Z_{\varphi(Q)} \end{pmatrix}$$

[4] D. Robert and N. S. "Halving Differential Additions on Kummer Lines". In: *EUROCRYPT 2025, Part VI*. vol. 15606, pp. 416–445

Half ladder (1/2)

$$n = 11 = \overline{1011}^2$$

$$\begin{tikzcd}
1 & 0_E \arrow[r, dash] \arrow[d] \arrow[dr] & P \arrow[d] \arrow[dl] \\
0 & P \arrow[r, dash] \arrow[d] \arrow[dr] & 2P \arrow[d] \\
1 & 2P \arrow[r, dash] \arrow[d] \arrow[dl] & 3P \arrow[d] \\
1 & 5P \arrow[r, dash] \arrow[d] \arrow[dl] & 6P \arrow[d] \\
 & 11P \arrow[r, dash] & 12P
\end{tikzcd}$$

Figure: Montgomery ladder

Half ladder (1/2)

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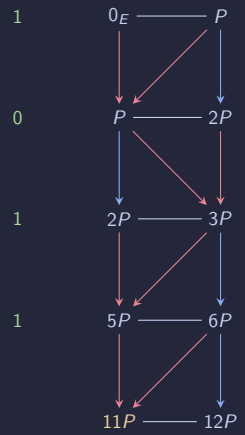

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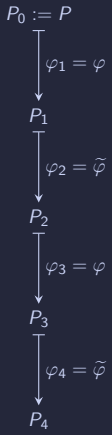

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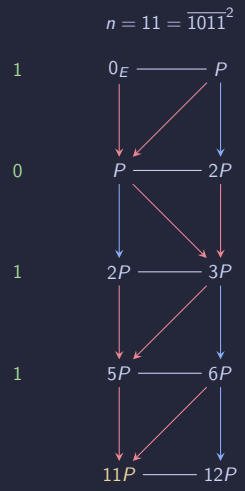

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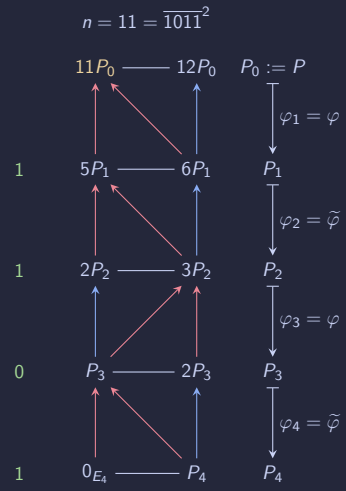

Figure: Half ladder

Algorithm 3: Half ladder step for a 2-isogeny φ

Input: $\varphi(R)$, $\varphi(S)$ where $R = m \cdot P$,
$S = (m+1) \cdot P$, b a bit

Output: $(2 \cdot R, R + S)$ if $b = 0$
$(R + S, 2 \cdot S)$ if $b = 1$

Data: The point P

```
def HalfxDBLADD_φ(φ(R), φ(S), b):
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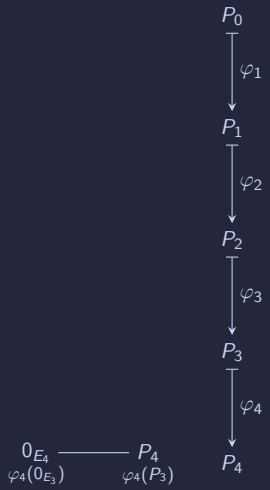

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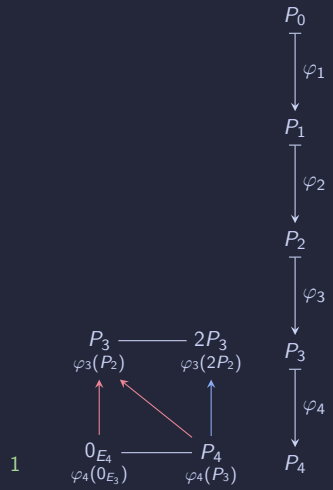

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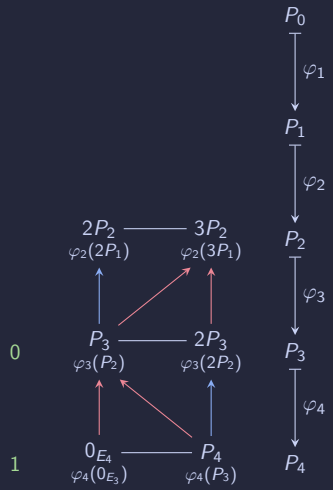

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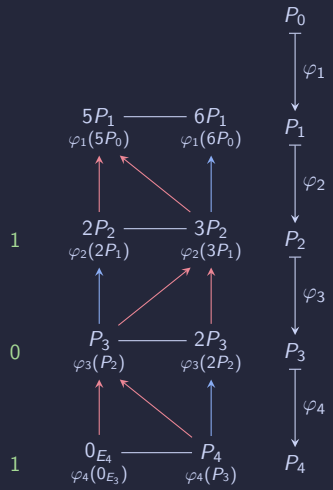

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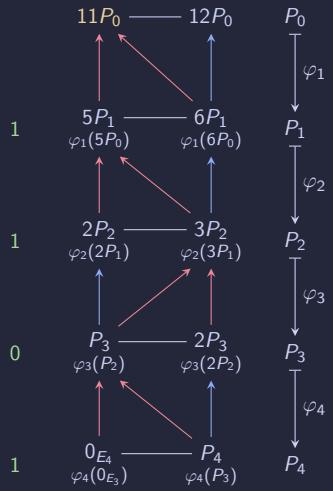

Figure: Half ladder

Computational cost: without pre-computation

On our theta model $\theta(a:b)$ previously studied, with $\varphi_{2i+1} = \varphi$ and $\varphi_{2i} = \widetilde{\varphi}$:

- $\varphi : \theta(a:b) \to \theta(A:B)$ and $\widetilde{\varphi}$: $2S + 1m_0$;
- $\mathtt{HalfDiffAdd}_{\varphi}$ and $\mathtt{HalfDiffAdd}_{\widetilde{\varphi}}$: $4M$;
- $\mathtt{HalfDouble}_{\varphi}$ and $\mathtt{HalfDouble}_{\widetilde{\varphi}}$: $2S + 1m_0$.

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- $\texttt{HalfDouble}_{\varphi}$ and $\texttt{HalfDouble}_{\widetilde{\varphi}}$: $2S + 1m_0$.

	Montgomery ladder	Half ladder, our contribution
Non-normalized base point	$6M + 4S + 1m_0$	$4M + 4S + 2m_0$
Normalized base point	$5M + 4S + 1m_0$ (or $4M + 4S + 2m_0$)	

Table: Ladder cost per bit with no pre-computation

Computational cost: with pre-computation

Algorithm	Pre-computation	Step
Montgomery ladder RtL[5]	$2M + 2S + 1m_0$	$4M + 2S$
Half ladder, our contribution	$2S + 1m_0$	$4M + 2S + 1m_0$

Table: Ladder costs per bit with a pre-computation but no normalization

[5] T. Oliveira et al. "How to (Pre-)Compute a Ladder - Improving the Performance of X25519 and X448". In: *SAC 2017*. Vol. 10719. Lecture Notes in Computer Science. Aug. 2017, pp. 172–191

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Half ladder, our contribution	$2S + 1m_0$	$4M + 2S + 1m_0$

Table: Ladder costs per bit with a pre-computation but no normalization

Also holds on a Montgomery curve with full 2-torsion (with a few tweaks)

$$\begin{cases} y^2 = x(x - A^2/B^2)(x - B^2/A^2), \\ \sqrt{\frac{A^2+B^2}{A^2-B^2}} \in k, \end{cases} \rightsquigarrow y^2 = x(x - a/b)(x - b/a).$$

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Conclusion

New ideas

- Framework for Kummer lines: flexibility;
- Variants of the Montgomery ladder;
- More general approach which could be used in higher dimension.

On-going and future projects

- Half ladder in dimension 2;
- Finding a Montgomery curve with small constants;
- Classification of Kummer lines with Galois representations;
- Having isogeny formulas in characteristic 2.